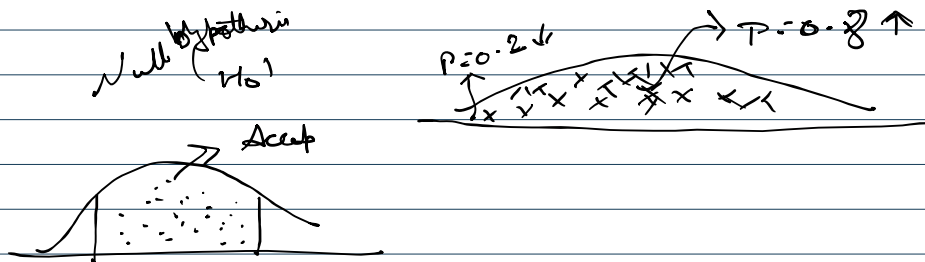


P-Value:-

The p-value is a number, calculated from a statistical test, that describes how likely you are to have found a particular set of observations if the null hypothesis were true. p-values are used in hypothesis testing to help decide whether to reject the null hypothesis or not.



$$S_0 \Rightarrow 0.5 \quad t_0 \Rightarrow 0.6 \\ 0-1$$

Hypothesis Test :- Eg :- Tossing a Coin for $n=100$ times

Probability of getting $P(H) \& P(T) \Rightarrow$ Fair $\Leftarrow P(H)=0.5 \& P(T)=0.5$

Fair $\Leftarrow P(H)=0.6 \& P(T)=0.4$

Semi fair

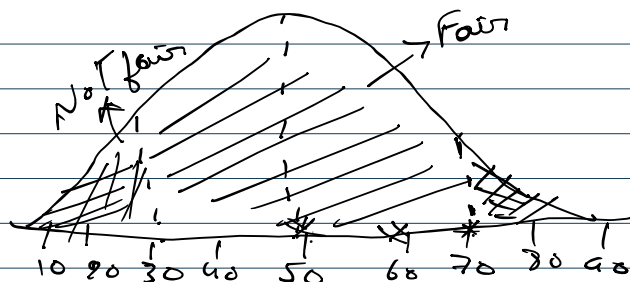
$\Leftarrow P(H)=0.7 \& P(T)=0.3$

① Null hypothesis
 $H_0 = \text{Coin is fair}$

② Alternate hypothesis
 $H_1 = \text{Coin is not fair}$

③ Experiment : 100 times

$$P(H)=50 \\ P(T)=50$$



$$P(H)=80 \quad P(T)=20 \quad \times$$

$$P(H)=60 \\ P(T)=40 \\ P(H)=70 \\ P(T)=30 \quad \cup$$

④ Significance Value :-

$$\alpha = 0.05 \quad (C.I.) \\ C.I = 1 - 0.05 \quad \alpha = 1 - C.I \\ = 0.95 \Rightarrow 95\%$$

$p < \text{Significance Value}$

Reject null hypothesis

Else

Fail to Reject Null hypothesis